

IFN647 Week 11 Review Questions

Question 1 (Distance Function)

Table 1 shows an example of a binary information table.

<i>Document</i>	t_1	t_2	t_3	t_4	t_5	t_6	t_7
d_1	1	1	0	0	0	0	0
d_2	0	0	1	1	0	1	0
d_3	0	0	1	1	1	1	0
d_4	0	0	1	1	1	1	0
d_5	1	1	0	0	0	1	1
d_6	1	1	0	0	0	1	1

Table 1. A binary information table

- (1) Draw the binary contingency table of Table 1 for comparing documents d_1 and d_2 .
- (2) Evaluate the Jaccard distance between d_1 and d_2 .
- (3) Evaluate the Simple matching coefficient distance between d_1 and d_2 .

Question 2 (Cluster Precision)

Table 2 shows a collection of documents, which includes 10 documents (each with a unique document id) $D = \{d_1, d_2, \dots, d_{10}\}$, two classes (spam and not spam) $Class = \{\text{'spam'}, \text{'not_spam'}\}$, and a vocabulary $V = \{w_1, w_2, \dots, w_5\} = \{\text{'cheep'}, \text{'buy'}, \text{'banking'}, \text{'dinner'}, \text{'the'}\}$.

document id	cheap	buy	banking	dinner	the	class
1	0	0	0	0	2	not spam
2	3	0	1	0	1	spam
3	0	0	0	0	1	not spam
4	2	0	3	0	2	spam
5	5	2	0	0	1	spam
6	0	0	1	0	1	not spam
7	0	1	1	0	1	not spam
8	0	0	0	0	1	not spam
9	0	0	0	0	1	not spam
10	1	1	0	1	2	not spam

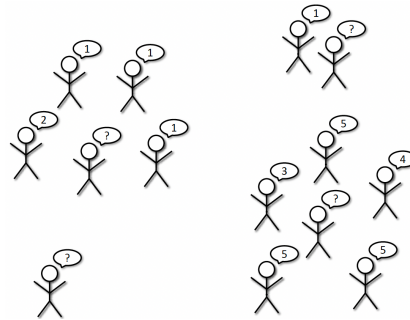
Table 2. A collection of documents

Assume your cluster algorithm produces two ($K=2$) clusters $C_1 = \{d_2, d_5, d_{10}\}$ and $C_2 = \{d_1, d_3, d_4, d_6, d_7, d_8, d_9\}$.

- (1) Calculate $MaxClass(C_i)$ for $i = 1$ and 2.
- (2) Calculate the cluster precision of your cluster algorithm.

Question 3 (Recommender systems)

Collaborative filtering leverages relationships between users to improve how items (documents) are matched to users (profiles). The figure below shows a group of users and their ratings for an item. The user with the question mark above its head has not rated this item yet. The goal of the recommendation algorithm is to fill in these question marks.



Suppose your team is designing functions to implement a **collaborative filtering–based recommender system**. For privacy preservation, all users and items are represented using anonymized identifiers:

- **Users:** represented by numeric IDs (e.g., 0, 1, 2) or symbolic IDs (e.g., u1, u2)
- **Items:** represented by numeric IDs or names
- **Ratings:** integers within a fixed scale (e.g., 0–5), where 0 indicates a missing (unobserved) rating

Example:

- A set of items:
 $I = [0, 1, 2, 3]$ (4 items)
- A set of users:
 $U = [0, 1, 2, 3, 4]$ (5 users)
- A user–item rating matrix represented as a list of user rating vectors:

```
Rvs = [  
  [1, 2, 1, 0],  
  [0, 1, 1, 1],  
  [2, 1, 3, 5],  
  [1, 0, 2, 0],  
  [0, 2, 3, 4]  
]
```

where each row corresponds to a user and each column corresponds to an item. A value of 0 denotes that the user has not rated the item.

Your task is to specify appropriate function names along with their corresponding input and output data structures for the following three functions:

- 1) `my_correlation(Rvs)`, returns users, items, and an user-user correlation matrix which measures user-user similarity.
- 2) `my_cluster(Rvs)`, returns a set of user clusters
- 3) `my_prediction(Rvs, C, U, I)`, returns the predicted rating to unseen items for all users

Question 4. (Recommender systems cont.)

Expand your solution to Question 3 by fully implementing the proposed functions in Python. This includes developing complete, well-documented implementations of the following components:

- **`my_correlation(Rvs)`**: compute the user–user similarity (e.g., Pearson correlation) and return a correlation matrix.
- **`my_cluster(Rvs)`**: group users into clusters based on their similarity (e.g., using k-means).
- **`my_prediction(Rvs, C, U, I)`**: predict ratings for unseen items (i.e., missing values) for all users using a collaborative filtering strategy.